

1. Write a Tcl script to print 'Hello, World!' on the screen.
2. Declare a variable called 'name' and assign your name to it. Print 'Welcome <name>'.
3. Write a script to take user input and display it back to the user.
4. Write a script to add two numbers entered by the user.
5. Perform subtraction, multiplication, and division of two numbers using 'expr'.
6. Write a script to check if a number is positive, negative, or zero.
7. Use an 'if-elseif-else' structure to determine whether a number is even or odd.
8. Create a 'for' loop to print numbers from 1 to 10.
9. Create a 'while' loop to print numbers from 10 to 1.
10. Create a list of 5 elements and print the entire list.
11. Access and print the 3rd element of a list.
12. Replace the second element of a list with a new value.
13. Sort a list and print the result.
14. Append an element to a list and print the updated list.
15. Remove an element from a list and display the list.
16. Write a procedure named 'greet' that takes a name and prints 'Hello, <name>'.
17. Write a procedure to calculate the square of a number.
18. Write a script to create and call a procedure that returns the maximum of two numbers.
19. Open a file 'data.txt' and write three lines of text to it.
20. Read the contents of 'data.txt' and print them line by line.
21. Count the number of lines in a file using Tcl.
22. Create a string and print its length.
23. Extract a substring from a given string.
24. Convert a string to uppercase (if manual logic required, use ASCII manipulation).
25. Compare two strings and print whether they are equal.
26. Split a string into a list using a space delimiter.
27. Join a list of strings into a single space-separated string.
28. Create an array with 5 key-value pairs and print all keys.
29. Print all values from a Tcl array.
30. Update the value of a key in a Tcl array.
31. Remove a key from a Tcl array and print the array.
32. Write a procedure to implement a simple calculator using switch-case style logic.
33. Write a script that checks whether a given year is a leap year.
34. Simulate a login prompt that checks username and password.
35. Create a menu system using 'switch' that performs different tasks based on user choice.
36. Use regular expressions to find digits in a string.
37. Validate an email address format using regex.
38. Write a script to reverse a string.
39. Write a procedure to compute the factorial of a number.
40. Implement a Fibonacci series generator using recursion.
41. Write a procedure that takes a list and returns the number of elements.
42. Write a script to demonstrate the use of 'catch' to handle errors.
43. Use 'after' to introduce a 2-second delay before displaying a message.

44. Create a GUI window with a button using Tcl/Tk (if Tk is available).
45. Add tooltips or labels to a GUI window in Tcl/Tk.
46. Create a nested list and access a sub-list element.
47. Sort a list of numbers in descending order.
48. Generate a random number between 1 and 100.
49. Write a procedure that takes a list and returns only even numbers.
50. Save the output of a script to a log file using redirection in Tcl.
51. Create a Tcl script that does the following:
 - Takes user input:** student name, age, and 3 subject marks.
 - Calculates:**
 - Total marks
 - Average marks
 - Whether the student passed (average ≥ 40)
 - Stores subjects** in a list and prints them.
 - Displays** all student details and results clearly.
52. Simulate a traffic light controller with a Tk GUI that changes colors based on a timer.
53. Create a procedure to merge and sort two large files line by line (like sort -m in UNIX).
54. Create a loop that finds all duplicate elements in a list and prints them only once.
55. Given a list of pairs (e.g., {{A 10} {B 20} {C 15}}), write a loop to find the key with the highest value.
56. Define a procedure that modifies a global variable without passing it as an argument.
57. Write a procedure to implement selection sort on a list.
58. Design a procedure that takes a file name and a search word, and returns line numbers where the word appears.
59. Write a procedure that finds the longest word in a list of strings.
60. Verify if a string starts with an uppercase letter.
61. Check if a string contains only digits.
62. Check if a string contains any special characters.
63. Extract phone numbers in format +91-0000000000
64. Read a file line-by-line, count the number of words in each line, and print the result
65. Build a small quiz game: ask 5 questions, take user input, and calculate score.
66. Write a script to remove duplicate lines from a file
67. Compare two files line-by-line and print lines that are different with line numbers.
68. Compare two lists and print elements that are in list A but not in list B.